



International Program in Artificial
Intelligence Engineering and
CyberSecurity

Lecture 01:

Introduction to cybercrime and security concepts

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Pre-Learning

The \$10 Trillion Problem

The Economic Magnitude

- Global Cost: Projected to exceed \$10.5 Trillion annually.
- The "Third Economy": If cybercrime were a nation, it would be the 3rd largest GDP in the world (after the U.S. and China).
- Beyond Stolen Cash: Costs include data destruction, lost productivity, theft of Intellectual Property (IP), and massive reputational damage.

Discussion: Why is Cybercrime More "Profitable" Than Bank Robbery?

Lesson Overview

We will define the legal and technical boundaries of cybercrime, explore the primary motivations of attackers, and introduce the "defender's bible": the **CIA Triad**. We wrap up by introducing the **Cyber Kill Chain** to understand how attacks are structured as a process, not a single event.

Lesson Learning Outcomes

By the end of this lecture, students will be able to:

- Classify criminal acts as either cyber-dependent or cyber-enabled.
- Evaluate the impact of a security incident using the CIA Triad framework.
- Identify the 7 stages of the Cyber Kill Chain.
- Recognize the different threat actor profiles (Script Kiddies, Hacktivists, Nation-States).

Keywords

- **CIA Triad:** Confidentiality, Integrity, Availability.
- **Cyber-Dependent:** Crimes that can only be committed using computers.
- **Threat Actor:** An individual or group that performs malicious actions.
- **Kill Chain:** A systematic process to plan and execute an attack.
- **Zero-Day:** A software vulnerability that is exploited before the developer knows about it.

Pre-Quiz

True or False:

A bank robbery committed via the internet is called a "Cyber-Enabled" crime.



Pre-Quiz

Which part of the CIA Triad is damaged if an attacker changes your bank account balance?

Answer:

Content

1.1 Defining the Landscape

1.2 The CIA Triad

1.3 Threat Actor Taxonomy

1.4 The Cyber Kill Chain



1.1 Defining the Landscape

The history

The general notion encircling hacking is that it started a few decades ago.

1830s – First Recorded “Hack”, the first data breach

- Based on optical telegraph system in France
- Used for transmitting government data between cities

Attack by Blanc Brothers (1834)

- Manipulated messages by bribing operators
- Inserted hidden signals to gain financial advantage in stock trading



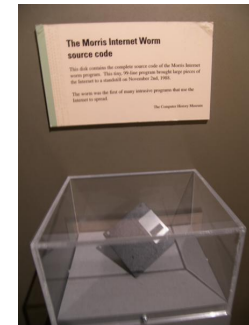
The history

The First Internet Attack – Morris Worm (1988)

The first major cyber incident was the Morris Worm, created by Robert Tappan Morris. His goal was not to cause harm but to measure the size of the internet. However, the program had a flaw—it kept replicating itself uncontrollably.

Impact:

- Spread across thousands of computers
- Affected about 10% of the internet at the time
- Caused major system slowdowns and crashes



The history

Cyber Warfare Era – Stuxnet (2010)

- Targeted Iranian nuclear facilities
- Attacked industrial control systems (ICS)
- Highly sophisticated and nation-state level

Significance:

- First cyber weapon targeting physical infrastructure
- Marked the beginning of cyber warfare

The history

Today's Threat Landscape

- Ransomware (encrypting data for money)
- Data theft and leaks
- Cryptocurrency mining malware

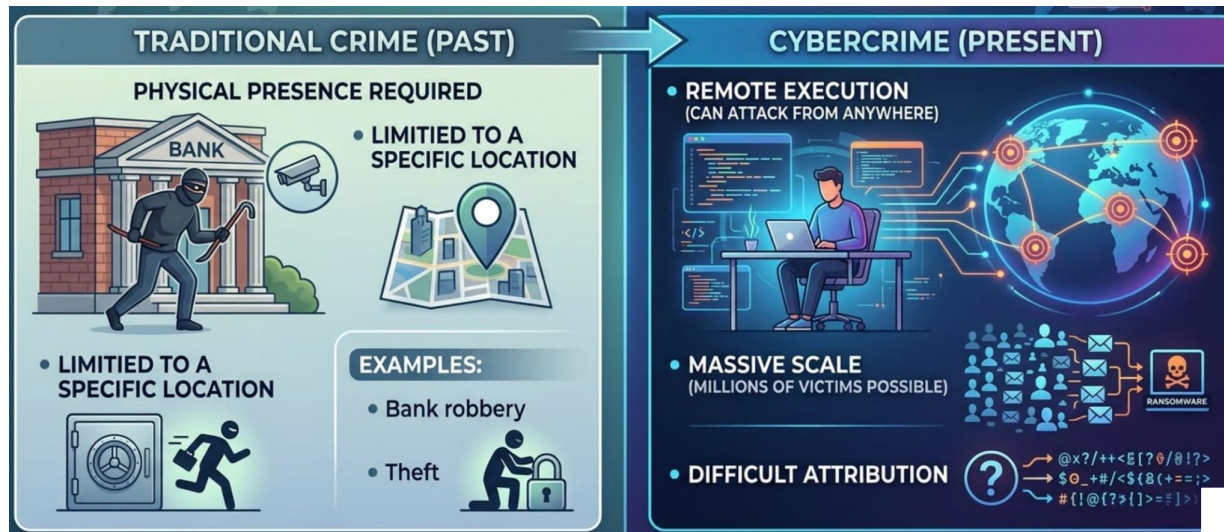
Key Trends:

- Larger attack surface (cloud, mobile, IoT)
- More sophisticated attackers
- Financial and political motivations



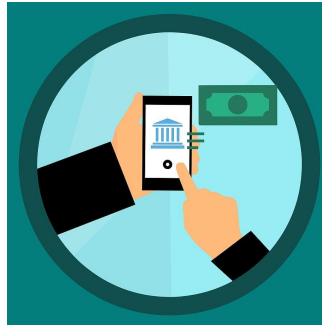
The Evolution of Crime

Moving from physical "smash and grab" to digital "click and disappear."



Why Cybercrime is Growing

- Increased internet usage
- Digital transformation (banking, healthcare, education)
- Anonymity of attackers
- Low cost, high reward



Real-World Case Study

WannaCry ransomware attack

- Affected 150+ countries
- Targeted hospitals, companies
- Encrypted files and demanded payment

Impact:

- UK hospitals canceled surgeries
- Financial losses in billions



Legal Frameworks

Challenges in Cyber Law:

- Crimes cross borders
- Different laws in different countries
- Difficult to identify attackers

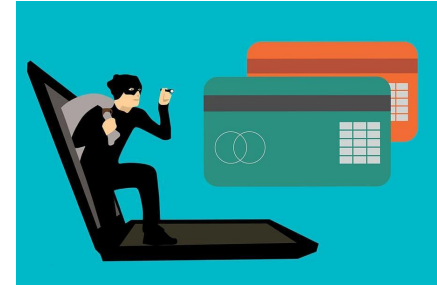


Budapest Convention

- Created by Council of Europe
- First international treaty on cybercrime

Key Objectives:

- Standardize cybercrime laws
- Enable international cooperation
- Improve investigation procedures



Cyber-Dependent vs Cyber-Enabled Crimes

Cyber-Dependent Crimes	Cyber-Enabled Crimes
Require computers/networks	Traditional crimes enhanced by technology
Cannot exist without technology	
Examples: • DDoS attacks • Malware • Ransomware	Examples: • Phishing scams • Identity theft • Cyberstalking



1.2 The CIA Triad

Overview

Three Pillars of Cybersecurity

- **Confidentiality**
- **Integrity**
- **Availability**



Why CIA Triad Matters

It is widely used in:

- System design
- Security policies
- Risk analysis

Example:

Bank system must:

- (Confidentiality):
- (Integrity):
- (Availability):

Open Discussion

The Scenario: "The Remote Wipe"

You lose your smartphone at a busy train station. You have a "Remote Wipe" feature enabled that allows you to delete all data from a web browser.

Could we comply all CIA in this scenario?



1.3 Threat Actor Taxonomy

What is a Threat Actor?

A threat actor is any individual, group, or organization that conducts a cyberattack or has the potential to exploit vulnerabilities in information systems, networks, or digital assets.



Comparison Table

Threat actors are defined not just by their technical ability but also by their intent, capability, and targets.

Actor	Skill	Motivation	Risk
Script Kiddies	Low	Fun	Low
Hacktivists	Medium	Ideology	Medium
Cyber Criminals	High	Money	High
Nation-State	Very High	Power	Very High

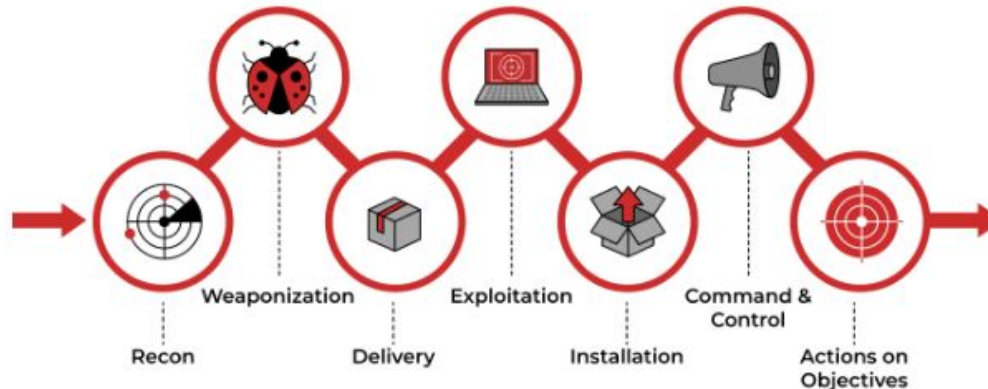


1.4 The Cyber Kill Chain

Overview

The cyber kill chain was developed by [Lockheed Martin](#). It is a framework describing the stages of a cyber attack

Goal: Understand attack flow to **anticipate, detect, and prevent threats**



Why It Matters

It helps defenders detect attacks early.

- Each stage = opportunity to block or mitigate damage

Framework can be applied to:

- Network defense
- Incident response
- Threat intelligence

Example Discussion

If a company monitors suspicious emails, which stage of the kill chain could be stopped?

Stage 1 – Reconnaissance

Definition: Attackers gather information about the target before attacking

Methods:

- Social media (LinkedIn, Facebook)
- Company websites
- Public records & databases
- Network scanning tools



Stage 2 – Weaponization

Definition: Crafting a weapon (malware) to exploit the identified vulnerabilities

Techniques:

- Embedding malware in files (PDF, Word, Excel)
- Exploit kits
- Trojan viruses

Stage 3 – Delivery

Definition: Sending the malware to the target system

Common Delivery Methods:

- Email attachments (phishing)
- USB drives / removable media
- Malicious links (drive-by downloads)

Stage 4 – Exploitation

Definition: Malware executes on the target system, exploiting vulnerabilities

How it works:

- User opens infected file
- Malware uses a software flaw to gain access
- Can escalate privileges to control system

Stage 5 – Installation

Definition: Malware installs itself to maintain persistent access

Techniques:

- Creating startup entries or services
- Injecting into system processes
- Rootkits to hide presence

Stage 6 – Command & Control (C2)

Definition: Attacker communicates with the infected system to control it remotely

Methods:

- HTTP/HTTPS traffic to attacker server
- Custom communication protocols
- Encrypted channels

Stage 7 – Actions on Objectives

Definition: The attacker achieves their ultimate goal

Common Objectives:

- Data theft → stealing sensitive info (financial, personal, intellectual property)
- Ransomware → encrypting files and demanding payment
- Destruction → sabotaging systems or physical infrastructure

Discussion Question

“If you could only defend against one stage of the kill chain, which would it be and why?”

Summary

- Cyber attacks are structured and sequential
- Each stage = an opportunity to detect and block the attack
- Defense strategy: disrupt one or more stages to prevent damage

Lecture Summary

Cybercrime is a multi-layered global threat. As security professionals, we don't just "fix bugs"—we defend the CIA Triad against specific Threat Actors. By understanding the Cyber Kill Chain, we learn that we don't have to be perfect; we just need to break the chain at one point to stop an attack.

Post-Quiz

1. An attacker shuts down a power grid. Which CIA pillar is the primary target?
2. In the Kill Chain, what is the difference between "Delivery" and "Exploitation"?
3. Why is the "Insider Threat" often more dangerous than an external Nation-State actor?

Reference

Title	Author(s)	ISBN	Year
Cybersecurity: The Beginner's Guide: A comprehensive guide to getting started in cybersecurity	Dr. Erdal Ozkaya	978-1789616194	2019

<https://medium.com/@kumarishefu.4507/try-hack-me-cyber-kill-chain-write-up-d076d323fdc8>

<https://www.lockheedmartin.com/en-us/capabilities/cyber/cyber-kill-chain.html>

THANK YOU!